When Retrieval Quality Decouples from Faithfulness: A Pre-Registered Audit of RAG Evaluation

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Submission under double-blind review

Abstract

RAG faithfulness claims are often reported from one retriever, one generator, one scorer, one threshold, and one dataset. We show that such claims are under-identified: the same fixed generations can support different reported contrasts depending on which reasonable audit axis is exposed. In 7,500 fixed Mistral generations, one intervention has raw score contrasts 0.011, 0.032, and 0.140 under DeBERTa, a second NLI-style proxy, and a local RAGAS-style judge; after z-scoring each scorer over the same rows, the contrasts remain separated (0.071, 0.231, 0.337 SD). A compact n=100 Qwen2.5 replication preserves the raw-contrast ordering (0.015, 0.046, 0.134). A 99-example two-rater human calibration reaches Cohen's $\kappa = 0.774$, while scorer alignment with the reference human label set is uneven (Spearman 0.140/0.380/0.441) and not settled. The proposed CCS explanation also does not survive a matched context test: in 200 HIGH/LOW CCS pairs at fixed mean query–passage similarity, HIGH–LOW faithfulness is -0.002 ($p = 0.628$, 95% CI $[-0.022, 0.017]$), while answer-span presence and CCS carry complementary diagnostic signal. At scale, the SQuAD/Mistral baseline / HCPC-v1 / HCPC-v2 cell shrinks to 0.661/0.650/0.661 over 2,500 examples per condition; a 5×5 threshold-transfer table has sign-changing cells; and a cost-aware head-to-head across CRAG, HCPC-v2, and RAPTOR-2L produces no dominating system. We distill these results into **ControlledRAG**, a reusable audit protocol and prescriptive minimum reporting standard whose seven axes are generator, retriever, context structure, scorer, human calibration, threshold transfer, and cost. The anonymized artifact contains auditable per-query outputs, matched context pairs, source tables, scripts, and supplement traces. ControlledRAG turns RAG faithfulness from a single-score claim into an auditable, seven-axis evidence standard.

1 Introduction

Retrieval-augmented generation (RAG) [Lewis et al., 2020] now powers deployed systems across medical QA, legal research, and customer support, yet many papers still evaluate faithfulness with a narrow template: one retriever, one generator, one scorer, one threshold, and one benchmark. That template is convenient, but it *under-identifies* the claim. A result that looks decisive in one generator/retriever/scorer cell can change sign, shrink, or become a deployment tradeoff under an equally defensible audit choice.

The trigger for this paper is a refinement-paradox result from our own prior work: improving per-passage retrieval appeared to reduce answer faithfulness. The headline was interesting enough to audit, but it does not survive the controls reported here. At scale, the original SQuAD/Mistral contrast shrinks to 0.661/0.650/0.661 across baseline / HCPC-v1 / HCPC-v2; a matched-similarity CCS intervention is null; threshold-transfer cells change sign; and a cost-aware head-to-head produces no dominating system.

The audit reframes the original headline as a stronger methodological result. The problem is not that a coherence hypothesis "failed"; the problem is that single-cell RAG faithfulness numbers hide too many degrees of freedom. Under fixed Mistral generations, three plausible scorers produce raw score contrasts 0.011, 0.032, and 0.140; z-scored contrasts on the same rows remain separated (0.071, 0.231, 0.337 SD); a compact Qwen2.5 replication preserves the raw-contrast ordering; and a two-rater human calibration shows that no automated scorer should be treated as an unqualified reference standard. The empirical results are not independent anecdotes: each one corresponds to an axis that a RAG faithfulness claim must disclose.

The positive contribution is **ControlledRAG**: a reusable audit protocol and prescriptive minimum reporting standard for RAG faithfulness claims. ControlledRAG is not a new retriever, not a new faithfulness metric, and not a method leaderboard. It asks authors to vary or explicitly hold fixed seven axes—generator,

retriever, context structure, scorer, human calibration, threshold transfer, and cost—so reviewers can tell whether a claim is robust or merely a metric, retriever, generator, threshold, or cost artifact.

Recommended minimum reporting standard for RAG faithfulness claims.

1. **Generator.** Family, parameter count, decoding settings, and whether outputs are fixed before scoring.
2. **Retriever.** Embedding model, indexer, reranker (if any), and any post-retrieval refinement step.
3. **Context-set structure.** At least two diagnostics beyond per-passage similarity (e.g., CCS, answer-span presence, mean query similarity).
4. **Scorer.** At least two faithfulness scorers, with a reported Pearson or Spearman cross-scorer correlation.
5. **Human calibration.** A multi-rater panel with inter-rater agreement and Spearman correlation against each automated scorer.
6. **Threshold transfer.** Sensitivity of any tuned threshold τ across datasets, reported as a transfer table.
7. **Cost.** Per-condition latency, indexing time, and external-call cost—including baselines.

Contributions. We make five contributions, each mapped to the standard.

1. **ControlledRAG standard (§4).** A seven-axis minimum reporting standard for RAG faithfulness claims.
2. **Fixed-generation metric fragility (§7).** A 7,500-row Mistral audit, a compact Qwen2.5 replication, and a 99-item human calibration showing why scorer choice is an audit axis rather than an implementation detail.
3. **Matched-context diagnostics (§5).** A pre-registered CCS null plus an answer-span/control diagnostic showing that answer-bearing evidence and context structure carry complementary signal in the matched cell.
4. **Transfer and deployment audit (§9–§11).** Threshold-transfer, noise, cost-aware baseline, stronger-retriever, and long-form stress results that expose the remaining audit axes without turning the paper into a method leaderboard.
5. **Review artifact.** An anonymized package containing auditable per-query outputs, matched context pairs, source tables, scripts, supplement traces, and documentation is provided for review; public release links will follow acceptance.

Roadmap. §2 positions the work against prior RAG evaluation. §3 describes the pipeline and the audited cells. §4 defines ControlledRAG. §5–§11 run the seven-axis audit. §12 discusses what survives and what does not. §13 returns to the broader literature. §14 states limitations and ethical considerations. §15 concludes.

2 Background and Related Work

The standard RAG pipeline. A modern RAG system [Lewis et al., 2020, Gao et al., 2024] threads a dense retriever [Karpukhin et al., 2020, Wang et al., 2024] returning the top-k passages by query–passage similarity, an optional cross-encoder reranker [Nogueira and Cho, 2019], an optional refinement module that further filters or compresses each retained passage (e.g., extractive sentence selection, query-conditioned summarisation, or learned compression [Xu et al., 2024]), and a generator that produces the final answer conditioned on the assembled context. Each stage individually is reported to improve task performance on standard benchmarks [Karpukhin et al., 2020, Izacard et al., 2022, Wang et al., 2024]; the implicit assumption is that better local retrieval produces more faithful answers globally.

Decoder-side correction. Two influential lines move the correction *into* the decoder. Self-RAG [Asai et al., 2024] fine-tunes the generator to emit "reflection tokens" that decide whether to retrieve, retrieve again, or stop, paying a fine-tune-time cost and a 2–3× inference latency cost. CRAG [Yan et al., 2024] introduces a lightweight retrieval evaluator that triggers web search when document relevance is judged poor, paying an external-call latency cost. Both take the position that the generator (or a learned auxiliary) is the right place to fix retrieval failures. ControlledRAG is orthogonal: it does not prescribe a place to fix retrieval and does not declare a winner among deployment policies. It prescribes how a faithfulness claim should be *reported* so that a reader can tell from the paper which deployment policy would actually have looked better under a different scorer or retriever.

Hierarchical retrieval. RAPTOR [Sarthi et al., 2024] clusters retrieved passages into a tree and retrieves from both leaves and LLM-summarised internal nodes, producing higher-coherence summary contexts at the cost of an offline LLM-summarisation pass. We compare against a 2-level RAPTOR baseline (§11) and find no single dominating system across faithfulness, hallucination, and latency.

Faithfulness evaluation. Faithfulness—whether a generated answer is supported by the retrieved context—is typically assessed via NLI-style scoring or LLM judging of the answer against the retrieved documents [Honovich et al., 2022, Laban et al., 2022, Es et al., 2024], occasionally complemented by human annotation. Our frozen artifact uses three automated support proxies: a DeBERTa-v3 zero-shot entailment label score (`cross-encoder/nli-deberta-v3-base`), a `roberta-large-mnli` zero-shot proxy, and a context-conditioned RAGAS-style local judge. The supplement reports the exact scorer input formats. Cross-scorer disagreement on the same fixed generations is itself one of the strongest pieces of evidence for ControlledRAG (§7).

RAG evaluation frameworks. Recent systems such as ARES [Saad-Falcon et al., 2024], RAGChecker [Ru et al., 2024], RAGTruth [Niu et al., 2024], MIRAGE [Park et al., 2025], FaithJudge [Tamber et al., 2025], RAGAS [Es et al., 2024], TRUE [Honovich et al., 2022], and SummaC [Laban et al., 2022] provide evaluators, datasets, benchmark suites, or diagnostic metrics. ControlledRAG is complementary. It does not replace those metrics or leaderboards; it asks whether a RAG faithfulness claim survives the hidden degrees of freedom those tools often instantiate.

Coherence and context structure. Discourse coherence has a long history in classical text generation [Li and Hovy, 2014, Mesgar and Strube, 2018, Nguyen and Karypis, 2021] but, to our knowledge, has not been quantified as a deployment-time RAG signal. Shi et al. [2023] show that LLMs are distracted by irrelevant context but do not consider the case where each individual passage is relevant yet the *set* is internally fragmented; Liu et al. [2024] document position-dependent context utilisation in long contexts. We treat the Context Coherence Score (CCS) and answer-bearing evidence presence as two complementary context-structure variables to *report* alongside per-passage similarity, not as substitutes for human evaluation or for cross-scorer checks.

What we add. Existing work proposes new retrievers, generators, scorers, datasets, or auxiliary models. We do not. ControlledRAG is not another metric suite or leaderboard. It is a disclosure standard: it asks authors to show which generator, retriever, context-structure, scorer, threshold, human-calibration, and cost choices their claim survives.

3 Setup and Audited Cells

We hold the entire RAG pipeline fixed across audit experiments and vary only the audit axis under test. The audited cells are deliberately narrow—primarily short-answer extractive QA on SQuAD and PubMedQA with 7B-class generators—so that scorer, retriever, and generator choices can be cleanly separated.

Datasets. The matched-context, scaled, metric-fragility, and span-control experiments use SQuAD [Rajpurkar et al., 2016]; the threshold-transfer and noise experiments additionally use PubMedQA [Jin et al., 2019], HotpotQA [Yang et al., 2018], NaturalQuestions [Kwiatkowski et al., 2019], and TriviaQA. The long-form stress test on QASPER and MS-MARCO is supplement-only and not treated as broad generalization evidence (§14 and supplement).

Pipeline. Embeddings come from `sentence-transformers/all-MiniLM-L6-v2`; indexing uses ChromaDB; refinement uses `cross-encoder/ms-marco-MiniLM-L-6-v2`. Default chunk size is 1024 characters with 10% overlap and default top-k is 3. Generators are served via local Ollama (Mistral-7B, Qwen2.5-7B). The matched-context audit also re-encodes the same 200 pairs with BGE-large and E5-large.

Conditions. Three retrieval configurations are evaluated head-to-head per query: **baseline** (top-k dense retrieval, no refinement), **HCPC-v**1 (Hybrid Context-Preserving Chunking: refine all retrieved passages through sub-chunking and cross-encoder selection—the original failure-mode condition), and **HCPC-v**2 (a coherence-gated HCPC variant with top-rank protection, a refinement cap, and merge-back). The audit treats these three conditions as instruments to measure the audit surface, not as systems to crown a winner.

Metrics. Per-query **faithfulness** in the legacy audit tables means the stored automated support score in $[0, 1]$; **hallucination** is the indicator that this score is < 0.5 unless a table explicitly states another label source. The frozen DeBERTa score averages sentence-level `cross-encoder/nli-deberta-v3-base` zero-shot entailment-label scores over generated answer sentences. The second automated score is a `roberta-large-mnli` zero-shot entailment-label proxy. In both legacy NLI-proxy implementations, the answer sentence is the sequence classified against the labels entailment/neutral/contradiction with `hypothesis_template="{}"`; the retrieved context remains in the row but is not consumed by that proxy. The RAGAS-style judge is context-conditioned: local `Mistral-7B` receives the question, retrieved context, and answer and returns a support score on $[0, 1]$ with a short rationale. This implementation detail is precisely why scorer choice is an audit axis rather than a hidden detail. **Retrieval similarity** is the mean cosine similarity between the query and each retained chunk. **CCS** is the mean minus the standard deviation of off-diagonal pairwise cosine similarities between retained chunk embeddings under `sentence-transformers/all-MiniLM-L6-v2`. The answer-span indicator is whether any normalised gold-answer alias appears in the retrieved context (lowercased, punctuation-stripped, whitespace-normalised lexical match, no model).

Statistical protocol. The matched-context test is pre-registered with paired Wilcoxon and a 10,000-resample bootstrap CI on the mean paired difference; the scaled audit reports seed-level paired Wilcoxon over five seeds; binary rates are reported with Wilson 95% CIs; the answer-span/control models report bootstrap 95% CIs on R^2 and AUROC. P-values outside the pre-registered primary matched tests are descriptive across multiple audit cells; we do not treat exploratory $p < 0.05$ rows as standalone discoveries. All audit cells, paired contrasts, and per-query CSVs are included in the anonymized review artifact.

4 ControlledRAG: The Minimum Reporting Standard

The seven-axis executive summary appeared in §1. This section turns it into a reporting standard. For each axis, the author must report what was varied or held fixed, why the axis matters, and which overclaim the disclosure prevents.

1. **Generator.** Report the generator family, parameter count, decoding settings, and whether outputs are fixed before scoring or regenerated per condition. This prevents Mistral-only conclusions from being read as generator-general claims; the Qwen2.5 replication is a compact example, not a complete generator panel.

2. **Retriever.** Report the embedding model, indexer, reranker, and any post-retrieval refinement step. This prevents MiniLM/BGE/E5 differences from being hidden under the word "retrieval"; stronger embeddings can change the retrieval surface (§5 and supplement).

Table 1: ControlledRAG seven-axis audit map. The table is a reporting standard, not a method prescription.

Axis	What to report	Why it matters	Evidence here	Overclaim prevented
Generator	model family, size, decoding, fixed vs regenerated outputs	same context can yield generator-specific behavior	Mistral main audit; Qwen2.5 compact replication	generator-universal claims
Retriever	embedder, index, reranker, refinement policy	retriever identity changes the context surface	MiniLM plus BGE/E5/GTE sanity checks	"retrieval quality" as one variable
Context structure	CCS, answer span, similarity, retained chunks	support and coherence can diverge	matched CCS null; span/control diagnostic	CCS as oracle or causal mediator
Scorer	scorer names, input format, thresholds, correlations	same generations can produce scorer-dependent contrasts	three-scorer fixed-output audit	single-score faithfulness claims
Human calibration	label protocol, raters, agreement, scorer alignment	automated scores are not reference labels	$n = 99$ two-rater calibration	metric ranking as settled
Threshold	tuned τ, tune data, off-diagonal transfer	tuned policies may not transport	5×5 transfer table	reusable threshold claims
Cost	latency, indexing time, external-call assumptions, hardware	deployment rankings depend on resource costs	CRAG / HCPC-v2 / RAPTOR-2L	leaderboard without cost

3. **Context structure.** Report at least two context-set diagnostics beyond per-passage similarity. CCS, mean query similarity, and answer-span presence are jointly informative (§5); this prevents coherence, similarity, and answer support from being collapsed into one retrieval quality number.

4. **Scorer.** Report at least two faithfulness scorers, their input format, and a Pearson or Spearman cross-scorer correlation. Raw score contrasts and standardized contrasts are scorer-sensitive on the same fixed generations (§7); this prevents single-metric faithfulness claims.

5. **Human calibration.** Report a multi-rater reference human label set, inter-rater agreement, and scorer correlation against the reference labels (§8). Calibration is not scorer endorsement; it prevents automatic metrics from being treated as reference labels.

6. **Threshold transfer.** Report how any tuned threshold τ transfers across datasets. Cross-dataset transfer is uneven (§9); this prevents a tuned threshold from being presented as a reusable policy without off-diagonal evidence.

7. **Cost.** Report latency, indexing time, and external-call cost for every condition, including baselines. This prevents method rankings that ignore deployment cost (§11).

The contribution is the standard, not any specific number. The rest of the paper supplies one matched, pre-registered self-application in which each empirical result motivates a required disclosure axis.

5 The Matched-Context Audit

The matched-context audit asks the diagnostic question directly: *at fixed mean per-passage query–passage similarity, does higher CCS predict higher faithfulness?* The null answer is not the end of the analysis. It creates a constructive diagnostic question: what variables distinguish answer support once mean similarity is fixed?

Construction. For SQuAD with Mistral, we form 200 HIGH/LOW pairs from the per-query audit data so that mean query–passage similarity is matched to within 0.02 between members of a pair while CCS is large in HIGH and small in LOW. Mean similarity is 0.4236/0.4215 across HIGH/LOW; mean CCS is 0.670/0.137. The similarity Wilcoxon is significant only because the residual noise is two orders of magnitude smaller than the CCS gap, not because the similarity gap is meaningful in absolute terms.

Result. Table 2 reports the pre-registered HIGH–LOW contrast. The mean paired difference is -0.002, paired Wilcoxon $p = 0.628$, Cohen's $d_z = -0.017$, 10,000-resample bootstrap 95% CI $[-0.022, +0.017]$. The HIGH hallucination rate is also *higher* than the LOW rate (16.5% vs. 9.0%). The pre-registered H_1 (HIGH>LOW faithfulness) is not supported in the audited cell. The continuous mean contrast and the

thresholded hallucination contrast measure different parts of the score distribution, so a null mean can coexist with threshold-sensitive tail differences. The matched test rules out the simple HIGH-CCS-improves-faithfulness story in this audited cell.

Table 2: Matched-context audit on SQuAD/Mistral (n=200 pairs, matched mean query–passage similarity).

Quantity	HIGH-CCS	LOW-CCS	Difference	Test
Mean similarity	0.4236	0.4215	+0.0021	matched by construction
Mean CCS	0.670	0.137	+0.533	—
Faithfulness	0.636	0.639	−0.002	paired Wilcoxon $p = 0.628$
Hallucination	16.5%	9.0%	+7.5 pp	McNemar $p = 0.011$
95% CI on faith diff	—	—	$[-0.022, +0.017]$	10,000-resample bootstrap

Span presence in matched contexts. The first candidate is answer-bearing evidence. The lexical span detector finds the gold answer string more often in LOW-CCS contexts than in HIGH-CCS contexts (27.0% vs. 17.0%, McNemar $p = 0.011$). This is itself a useful finding: under the matched-similarity construction, the structural property captured by CCS is not the same property as "the retrieved context contains the answer". The lexical span variable is also a weak proxy: a context can contain the answer string while failing to support the generated answer, and thresholded hallucination and mean score can diverge. ControlledRAG does not discard context structure; it decomposes it.

Answer-span/control diagnostic. Table 3 extends the span check into a regression and classification panel on the matched cell ($n = 400$ contexts). We fit single-variable, two-variable, and three-variable models against (i) continuous DeBERTa faithfulness (R^2) and (ii) the binary hallucination indicator (AUROC).

Table 3: Answer-span/control diagnostic on the matched SQuAD/Mistral cell. R^2 is for continuous DeBERTa faithfulness; AUROC is for the binary hallucination label. Bootstrap 95% CIs are in the supplement.

Predictors	Faithfulness R^2	Hallucination AUROC
answer span present	0.019	0.531
mean query similarity	0.002	0.586
CCS	0.003	0.639
answer span + similarity	0.024	0.590
answer span + CCS	0.022	0.631
answer span + similarity + CCS	**0.025**	**0.643**

Interpretation. Three patterns are robust within the audited cell. (i) For continuous faithfulness, answer-span presence is the strongest *single* predictor. (ii) For binary hallucination, CCS is the strongest single predictor; the lexical span check is barely above chance. (iii) The full feature set has the highest score on both quantities. The R^2 values are small in absolute terms, so the table is not an explanatory model of faithfulness. Its value is comparative: answer support and context structure diagnose complementary failure modes, and a faithfulness claim is more informative when both are reported. We deliberately do not claim that either variable has a directional role, that span subsumes CCS, or that CCS subsumes span.

6 The Scaled Audit

The matched-context audit is intentionally local. The scaled audit asks the same headline question—does HCPC-v1 refinement reliably hurt faithfulness on SQuAD/Mistral?—at n=2,500 examples per condition with five seeds, with each seed analysed by paired Wilcoxon on 500-pair contrasts.

Result. Table 4 reports the headline table over the five seeds. The pooled means are 0.661/0.650/0.661 for baseline / HCPC-v1 / HCPC-v2, with overlapping Wilson CIs on hallucination rate (14.6%/14.7%/14.3%). Per-seed paired Wilcoxon yields one of five seeds significant at $p < 0.05$ for the baseline−HCPC-v1 contrast

(seed 44, paired $p = 0.008$); the other four seeds yield $p \in [0.083, 0.234]$. The v2−v1 contrast reaches significance in 3 of 5 seeds, but the magnitude is small (per-seed mean diff 0.003–0.019).

Table 4: Scaled audit on SQuAD/Mistral, $n = 2{,}500$ per condition, pooled across five seeds.

Condition	Faithfulness (95% CI)	Hallucination (Wilson)	Retrieval sim.	Refine rate
Baseline	0.661 [0.655, 0.667]	14.6% [13.3, 16.0]	0.532	0.000
HCPC-v1	0.650 [0.645, 0.656]	14.7% [13.4, 16.2]	0.569	1.000
HCPC-v2	0.661 [0.655, 0.667]	14.3% [13.0, 15.7]	0.532	0.705

Interpretation. The original $n = 30$ headline contrast ($0.7995 \rightarrow 0.6407$) does not survive at scale. The scaled contrast on SQuAD/Mistral is small and seed-sensitive, and the headline should not be cited as a broad refinement-hurts-faithfulness claim. This finding is itself a ControlledRAG-style audit lesson: at $n = 30$, the contrast looked clean enough to be reported as a phenomenon, but the contrast at $n = 2{,}500$ rejects that reading without further controls. We treat the remaining audits as the controls, not as reasons to reverse the original claim.

7 Fixed-Generation Metric Fragility

This section is the core empirical pillar for the scorer axis. For the scaled audit's 7,500 Mistral rows, outputs are fixed before rescoring, retrieval condition is fixed, and the only changed component is the scorer. The three scorers are not guaranteed to be calibrated to the same numerical scale, so raw score differences are not interpreted as directly comparable causal estimates. The important claim is that the sign, magnitude, and ranking of the reported contrast are scorer-sensitive even on fixed generations. We then replicate the raw contrast compactly on a second generator (Qwen2.5) at $n = 100$ fixed contexts per condition.

Table 5: Fixed-generation metric fragility. Raw values are baseline−HCPC-v1 score differences on each scorer's native scale. Mistral z/rank columns standardize each scorer over the same fixed rows before computing the paired contrast; bootstrap CIs are in the supplement.

Scorer	Mistral raw	Mistral z	Mistral rank	Qwen2.5 raw
DeBERTa proxy	0.011	0.071 SD	0.022	0.015
roberta-large-MNLI proxy	0.032	0.231 SD	0.062	0.046
RAGAS-style judge	0.140	0.337 SD	0.085	0.134

Table 6: Cross-scorer correlations on fixed-generation rows.

Metric pair	Mistral r	Mistral ρ	Qwen2.5 r	Qwen2.5 ρ
DeBERTa vs. roberta-large-MNLI	0.259	0.265	0.380	0.429
DeBERTa vs. RAGAS-style	0.182	0.212	0.280	0.318
roberta-large-MNLI vs. RAGAS-style	0.674	0.651	0.721	0.668

Scorer implementations. The DeBERTa and `roberta-large-mnli` rows are legacy zero-shot entailment-label proxies from the frozen artifact; the local RAGAS-style judge is the context-conditioned scorer. This is not a critique of any metric. It is the audit point: scorer model, prompt, input formatting, and threshold are part of the reported claim.

Interpretation. Scorer choice is, on this audit, the single largest movable quantity in the headline number. The human-calibration panel in the next section also shows that no scorer should be treated as an unqualified reference standard. Scorer choice is therefore an audit axis, not an implementation detail: ControlledRAG requires at least two scorers and a cross-scorer correlation to be reported. Scorer triangulation is not proof of correctness; it discloses how much of the reported number is scorer-dependent.

8 Human Calibration

Two raters (one expert, one general-domain) independently label 99 matched-cell items with a faithful/hallucinated decision and a 0–1 confidence score. We use rater 1's labels as the reference human label set and rater 2's labels as an independent reliability check; we do not collapse the two raters by majority or discussion (with two raters either procedure would be ill-defined). The supplement details the protocol.

Agreement and correlation. Raters reach Cohen's $\kappa = 0.774$ on the binary label, which is in the substantial- to high-agreement range for this kind of task. Spearman correlations between the rater-1 label set and the three automated scorers are $\rho_{\text{DeBERTa}} = 0.140$, $\rho_{\text{second NLI}} = 0.380$, and $\rho_{\text{RAGAS-style}} = 0.441$. Bootstrap 95% CIs on the three correlations overlap, so the metric ranking is suggestive rather than settled.

Interpretation. Two human raters who reach substantial agreement still give an uneven picture of scorer alignment. The right reading is calibration, not scorer endorsement: the panel reports how well each scorer aligns with the reference human label set on this audited cell, and that alignment is uneven enough to merit explicit disclosure. ControlledRAG asks for this calibration as a routine reporting axis, not as a means of crowning one scorer.

9 Threshold Transfer

A faithfulness claim that depends on a tuned operating threshold τ should report whether τ transfers across datasets. We sweep $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ for a CCS-gate retrieval intervention. For each tune dataset, τ^* is the threshold that maximises recovery on that dataset alone; diagonal recovery evaluates that τ^* on the same dataset, while off-diagonal recovery evaluates it on the other datasets. Recovery is a dimensionless normalised contrast,

$$R_\tau(d) = \frac{\text{faith}_{\text{gate},\tau}(d) - \text{faith}_{\text{HCPC-}v1}(d)}{\text{faith}_{\text{baseline}}(d) - \text{faith}_{\text{HCPC-}v1}(d)}.$$

Values can exceed 1 when the gate beats the baseline relative to HCPC-v1, and they can be negative or sign-inverted when the baseline−HCPC-v1 denominator is small or negative. We therefore use the table as a transport diagnostic, not as a universal utility score. A tuned threshold is not a reusable policy unless this off-diagonal transfer is reported.

Table 7: Threshold transfer summary. For each tune dataset, τ^* is the value of τ that maximises recovery on that dataset alone; "diagonal recovery" is recovery at τ^* on the same dataset; "off-diagonal mean" is the mean recovery at τ^* over the four *other* eval datasets. A large positive gap means the tuned τ^* does not transport. The full per-cell transfer matrix is in supplement §9.

τ tuned on	τ^*	Diagonal recovery	Off-diagonal mean	Gap (diag − off-diag)
PubMedQA	0.4	+1.452	+0.451	+1.002
NaturalQs	0.5	+1.401	+0.211	+1.191
SQuAD	0.3	+0.823	−0.141	+0.963
TriviaQA	0.4	+0.464	+0.698	−0.234
HotpotQA	0.3	−0.112	+0.093	−0.205

Interpretation. PubMedQA, NaturalQuestions, and SQuAD show a positive diagonal-vs-off-diagonal gap large enough to flag: tuning τ on one of them and transporting it loses most of the on-diagonal normalized contrast. TriviaQA and HotpotQA show a negative gap; their τ^* does not win on their own data (HotpotQA's diagonal value is itself negative), so transport is moot. The supplement reports the full 5×5 per-cell matrix. ControlledRAG asks for the transfer table to be reported because per-cell normalized contrasts can change sign depending on which dataset τ is tuned on; we do not claim universal threshold failure from this one sweep.

10 Coherence vs. Generic Retrieval Noise

A reasonable objection to any context-structure diagnostic is whether it is just a re-statement of generic retrieval noise. We test this by injecting controlled retrieval noise at three rates (1/3, 2/3, and fully replaced) and comparing two noise types: random off-topic passages, and coherence-preserving uninformative passages (passages drawn from the same topic distribution that do not contain the answer).

Result. The faithfulness slope per noise rate is -0.069 for random noise and -0.043 for coherence-preserving uninformative noise. The similarity slope is much larger for random noise (-0.481 per noise rate) than for coherence-preserving noise (-0.113), as expected. Coherence-preserving noise has a distinguishably smaller faithfulness drop at the same nominal substitution rate.

Interpretation. On this matched comparison, coherence carries some signal that is not captured by mean per-passage similarity. We do not claim that the gap is large in absolute terms, only that the two noise types produce distinguishable faithfulness slopes at matched substitution rate. The right reading is that coherence and similarity are correlated but not identical; ControlledRAG asks for both to be reported when a context-structure claim is made.

11 Cost-Aware Baselines

Faithfulness numbers reported without latency and indexing cost are under-identified along the cost axis. This section is not a leaderboard; it is an identifiability test for deployment claims. We re-run a head-to-head on $n = 200$ SQuAD and $n = 200$ HotpotQA queries with three baselines: CRAG [Yan et al., 2024], HCPC-v2, and a 2-level RAPTOR [Sarthi et al., 2024] index.

Table 8: Cost-aware head-to-head, $n = 200$ per (dataset, condition). Latency is end-to-end per query. Indexing time is the offline cost of building the dataset's retrieval structure. Faithfulness CIs are bootstrap 95% intervals; hallucination CIs are Wilson 95% intervals. RAPTOR-2L's indexing cost is 17–22× the dense baseline's.

Dataset	Condition	Faithfulness	Hallucination	Mean latency	Indexing
SQuAD	CRAG	0.698 [0.675, 0.722]	12.0% [8.2, 17.2]	1,440 ms	8.0 s
SQuAD	HCPC-v2	0.708 [0.685, 0.732]	12.5% [8.6, 17.8]	1,720 ms	8.0 s
SQuAD	RAPTOR-2L	0.710 [0.685, 0.735]	12.5% [8.6, 17.8]	1,714 ms	175.4 s
HotpotQA	CRAG	0.643 [0.627, 0.658]	10.5% [7.0, 15.5]	1,941 ms	8.0 s
HotpotQA	HCPC-v2	0.633 [0.617, 0.650]	12.5% [8.6, 17.8]	2,289 ms	8.0 s
HotpotQA	RAPTOR-2L	0.631 [0.615, 0.647]	13.0% [9.0, 18.4]	2,414 ms	134.0 s

Harness and cost accounting. This is a closed, zero-dollar local/free-tier harness: local Ollama/Mistral generation, Chroma indexing, MiniLM cross-encoders, and no paid APIs. The CRAG reimplementation uses the documented evaluator/strip-refinement structure with live web search disabled, so its external-call count and external dollar cost are both zero in this cell. A live-web CRAG run would need to report web-call count, API cost, and network latency as a separate deployment setting.

Self-RAG. Self-RAG [Asai et al., 2024] is reported in the supplement because its reflective decoding protocol violates the fixed-context assumption used in the main audit. ControlledRAG asks that any baseline of this kind be reported with its harness and excluded from head-to-head averages it cannot fairly support.

Interpretation. On SQuAD, all three systems are within 1.2 percentage points on faithfulness with overlapping intervals; HCPC-v2 and RAPTOR-2L have the same hallucination rate while CRAG is numerically lower. On HotpotQA, CRAG is numerically strongest on faithfulness, hallucination, and latency in this small cell. RAPTOR-2L's offline indexing time is 17–22× the dense baseline's. There is no method that is uniformly preferred once faithfulness, hallucination, latency, and indexing cost are considered, and ControlledRAG's cost axis exists precisely so that contrasts of this kind are reported rather than chosen.

12 Discussion

What survives. The under-identification thesis is not a collection of negative results; it is the reason a reporting standard is needed. The matched intervention is null, the scaled headline is small and seed-sensitive, fixed-generation metric fragility changes raw and standardized reported contrasts, the threshold table has sign-changing cells, and the cost-aware head-to-head has no dominator. Each cell corresponds to a ControlledRAG axis. The positive contribution is a standard of evidence: RAG faithfulness claims should not be accepted from a single retrieval score, faithfulness scorer, tuned threshold, or favorable baseline row.

What CCS is and is not. CCS is one of several context-set diagnostics ControlledRAG asks for. It is not a faithfulness oracle, not a guarantee, and not a substitute for human evaluation or for cross-scorer triangulation. The matched-context audit rules out a clean directional mediating role for CCS in the audited cell. The answer-span/control diagnostic shows that span and CCS are complementary axes, with span carrying more continuous-faithfulness signal and CCS carrying more hallucination-AUROC signal. We report both; ControlledRAG asks readers to do likewise.

Where the audit stops short. Two visible gaps remain. The second-generator replication uses one local Qwen2.5 model, not a broad generator panel; the long-form stress test on QASPER and MS-MARCO has only 40 distinct questions and is supplement-only. We report both gaps as part of the audit because the standard is meant to make scope visible, not to hide it.

13 Related Work, Continued

Beyond the RAG-evaluation context in §2, this paper sits next to a small but growing literature on RAG *evaluation methodology*. Honovich et al. [2022] re-evaluate factual-consistency evaluation broadly and conclude that no single NLI-based scorer is universally preferred; ControlledRAG's metric-fragility axis is the operational consequence of that conclusion for RAG. Es et al. [2024] propose RAGAS as a metric suite; we use a RAGAS-style judge as one of three scorers and treat its disagreement with NLI scorers as evidence for the standard, not as a critique of RAGAS.

Table 9: Positioning against recent RAG evaluation work.

Work	Primary contribution	ControlledRAG relation
ARES [Saad-Falcon et al., 2024]	trained judges for context relevance, answer faithfulness, and answer relevance with PPI calibration	complementary evaluator; ControlledRAG asks whether claims survive scorer/retriever/generator choices
RAGChecker [Ru et al., 2024]	fine-grained retrieval and generation diagnostics for RAG systems	complementary metric framework; ControlledRAG is the disclosure layer around such diagnostics
RAGTruth [Niu et al., 2024]	manually annotated hallucination corpus with word-level labels	complementary benchmark; ControlledRAG asks how benchmark conclusions transport across audit axes
MIRAGE [Park et al., 2025]	RAG benchmark with component-specific metrics and adaptability probes	complementary benchmark; ControlledRAG emphasizes reporting hidden degrees of freedom
FaithJudge [Tamber et al., 2025]	LLM-as-judge faithfulness benchmark and evolving leaderboard	complementary scorer/leaderboard; ControlledRAG requires scorer calibration and cross-scorer disclosure
RAGAS / TRUE / SummaC [Es et al., 2024, Honovich et al., 2022, Laban et al., 2022]	reference-free or NLI-style factual-consistency metrics	metrics used as audit inputs, not treated as reference standards

14 Limitations and Ethics

Audited surface. The audited cells are mostly short-answer extractive QA on SQuAD with 7B-class generators. The matched-context audit is on SQuAD only. The conclusions are scoped to that surface. Stronger embeddings (BGE-large, E5-large) preserve the matched HIGH>LOW CCS ordering but the

fixed-generation result remains null because the generations are unchanged; the supplement reports the multi-retriever generation table as a sanity check, not as exhaustive robustness.

Generators. The second-generator replication uses one local Qwen2.5 model at $n = 100$ fixed contexts per condition. The metric-fragility ordering is preserved, but ControlledRAG explicitly asks for a broader generator panel before any universal-generator claim can be made. We do not make that claim here.

Long-form scope. The QASPER/MS-MARCO long-form stress test has 40 distinct questions and is included only as an exploratory scope probe in the supplement. The short-answer paradox does not generalise cleanly to that small long-form sample (MS-MARCO span-faith paradox drop -0.033, QASPER 0.002). This is reported as a scope limit, not as broad long-form generalization evidence.

Human evaluation. Two raters at $n = 99$ is calibration, not a settled scorer ranking. Bootstrap CIs on the Spearman correlations overlap; the human ranking is suggestive only.

Self-RAG harness. The Self-RAG [Asai et al., 2024] appendix baseline runs in a harness mismatched with the audit's fixed-context protocol. We report the row in the supplement and exclude it from head-to-head averages. We explicitly do not generalize from this mismatched harness to Self-RAG overall.

Ethics. The audit uses public datasets and free-tier or local compute. No human-subjects data is collected beyond the two-rater calibration panel, which is not personally identifiable and does not involve sensitive content. The review artifact is anonymized; public release links will follow acceptance.

Scope as a feature. These limitations are precisely the axes ControlledRAG asks future papers to disclose. Our evidence is deliberately scoped, and the standard makes that scope explicit.

15 Conclusion

A RAG faithfulness claim should not be a single number tied to a single scorer, retriever, generator, threshold, or baseline. The audit experiments in this paper, taken together, show that under any one of those collapsed axes the reported headline can move materially. The positive contribution is a prescriptive minimum reporting standard, ControlledRAG, that names the seven axes a faithfulness claim has to report together, and demonstrates the standard by self-applying it to our own prior contribution. ControlledRAG turns RAG faithfulness evaluation from a single-score claim into a seven-axis auditable evidence standard.

References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*, 2024.

Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. RAGAS: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–163, 2024.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2024.

Or Honovich, Roee Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kukliansy, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. TRUE: Re-evaluating factual consistency evaluation. *NAACL*, 2022.

Gautier Izacard, Patrick Lewis, Maria Lomeli, Lucas Hosseini, Fabio Petroni, Timo Schick, Jane Dwivedi-Yu, Armand Joulin, Sebastian Riedel, and Edouard Grave. Few-shot learning with retrieval augmented language models. *arXiv preprint arXiv:2208.03299*, 2022.

Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 2567–2577, 2019.

Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergei Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6769–6781, 2020.

Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466, 2019.

Philippe Laban, Tobias Schnabel, Paul N. Bennett, and Marti A. Hearst. SummaC: Re-visiting NLI-based models for inconsistency detection in summarization. In *Transactions of the Association for Computational Linguistics (TACL)*, 2022.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.

Jiwei Li and Eduard Hovy. A model of coherence based on distributed sentence representation. *Proceedings of EMNLP*, 2014.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics (TACL)*, 2024.

Mohsen Mesgar and Michael Strube. A neural local coherence model for text quality assessment. *Proceedings of EMNLP*, 2018.

Dat Quoc Nguyen and George Karypis. Conditional discourse coherence modeling. *arXiv preprint arXiv:2106.04425*, 2021.

Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10862–10878, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.585. URL https://aclanthology.org/2024.acl-long.585/.

Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*, 2019.

Chanhee Park, Hyeonseok Moon, Chanjun Park, and Heuiseok Lim. MIRAGE: A metric-intensive benchmark for retrieval-augmented generation evaluation. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 2883–2900, Albuquerque, New Mexico, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-naacl.157. URL https://aclanthology.org/2025.findings-naacl.157/.

Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 2383–2392, 2016.

Dongyu Ru, Lin Qiu, Xiangkun Hu, Tianhang Zhang, Peng Shi, Shuaichen Chang, Jiayang Cheng, Cunxiang Wang, Shichao Sun, Huanyu Li, Zizhao Zhang, Binjie Wang, Jiarong Jiang, Tong He, Zhiguo Wang, Pengfei Liu, Yue Zhang, and Zheng Zhang. RAGChecker: A fine-grained framework for diagnosing retrieval-augmented generation. *arXiv preprint arXiv:2408.08067*, 2024. doi: 10.48550/arXiv.2408.08067. URL https://arxiv.org/abs/2408.08067.

Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. ARES: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 338–354, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.20. URL https://aclanthology.org/2024.naacl-long.20/.

Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. RAPTOR: Recursive abstractive processing for tree-organized retrieval. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2401.18059.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2023.

Manveer Singh Tamber, Forrest Sheng Bao, Chenyu Xu, Ge Luo, Suleman Kazi, Minseok Bae, Miaoran Li, Ofer Mendelevitch, Renyi Qu, and Jimmy Lin. Benchmarking LLM faithfulness in RAG with evolving leaderboards. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 799–811, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-industry.54. URL https://aclanthology.org/2025.emnlp-industry.54/.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. Improving text embeddings with large language models. *arXiv preprint arXiv:2401.00368*, 2024.

Fangyuan Xu, Weijia Shi, and Eunsol Choi. RECOMP: Improving retrieval-augmented LMs with compression and selective augmentation. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2310.04408.

Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 2564–2579, 2024.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380, 2018.